

Sensors and Sensing

Fall 2026

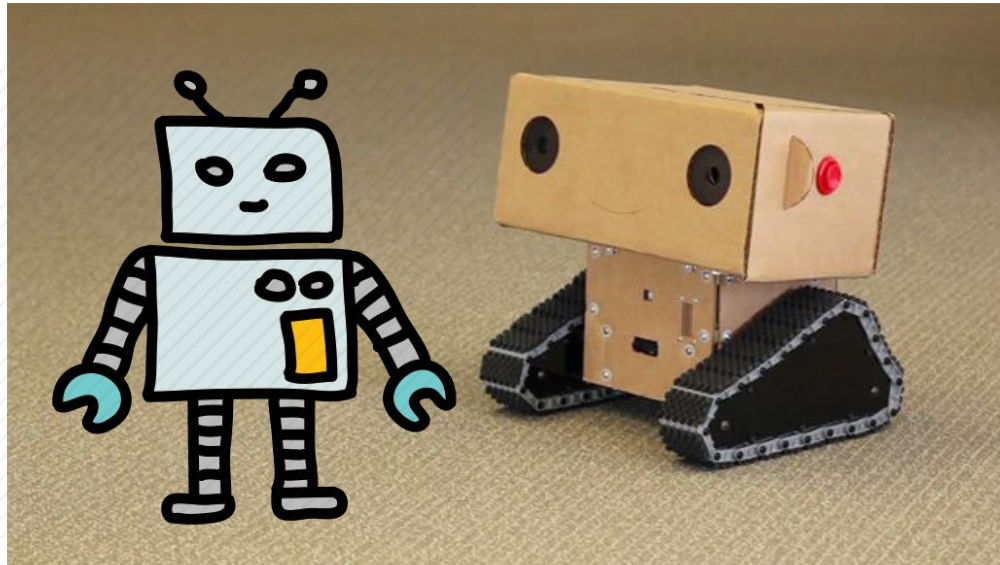
Quick Look at Sensors

Lots of sensor types:

- Proximity/Distance
 - Tactile
 - Ultrasonic
 - Capacitive
 - etc
- Orientation
- Location
- Temperature
- Sound Level
- Sensors on your robot

Candy Question!

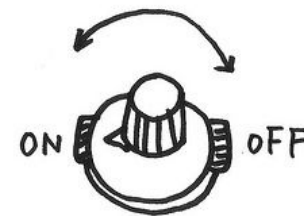
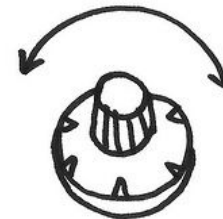
- What is the difference between a digital and analog sensor?
- Name an example of each!



Answer:

- A digital sensor returns a 0 or a 1
 - Example: bump sensor, button
- An analog sensor returns a value from a range based on the voltage level.
 - Examples: photocell, IR rangefinder

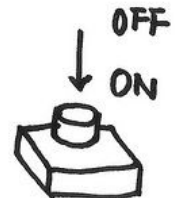
VOLUME knob



Toggle Switch



PUSHBUTTON



Sensors: Proximity: Reflective

- Camera sensors
 - Computer Vision
- Photoresistors
- Photodiodes
- Vary in price due to range of quality



Sensors: Proximity: Tactile

- Limit Switches
- On-Off Switch

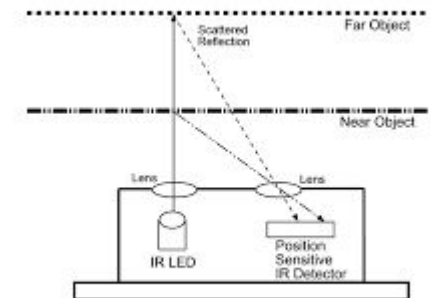
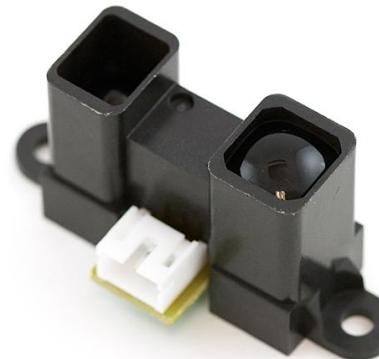
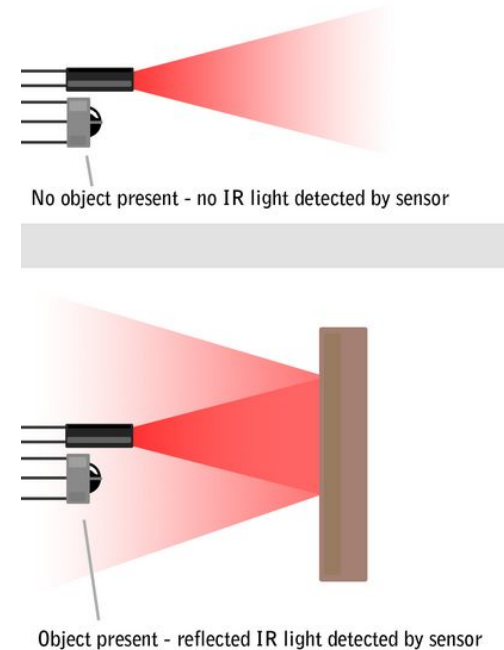
Input changes once the Switch is pressed.

Extremely easy to read/
either On or Off



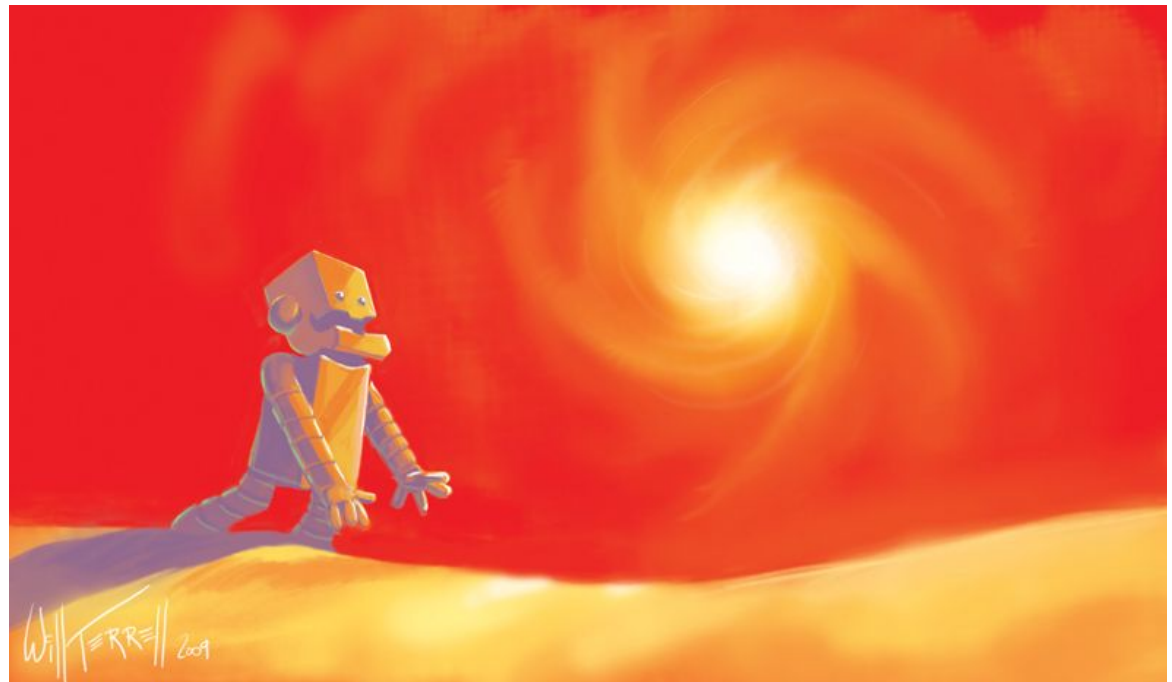
Sensors: Proximity: Infrared

- Bounces IR beam off target
- Cheap, reliable
- Analog, returns distance
- Range: 5 – 150 cm
- Some similar sensors use ultrasonic instead

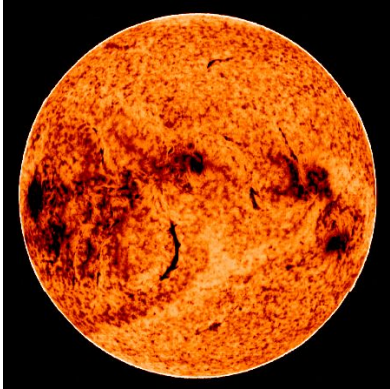


Candy Question!

- Do IR sensors work well in direct sunlight?
- Why don't we always use ultrasonic sensors?



Answer:



- IR sensors do not work well in direct sunlight because the light and heat from the sun interferes with the sensor readings. In other words, the sun emits IR radiation as well, interfering with the sensor.
- Multiple robots running ultrasonic sensors interfere with each other, since one robot may pick up another robot's soundwaves.

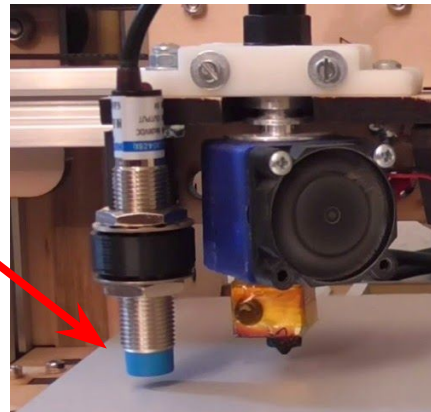
Sensors: Proximity: Capacitive

- Detects changes in capacitance
- (Like a touch screen)
- Requires target to be of somewhat conductive material
- Analog output
- Range: 2 – 40 mm



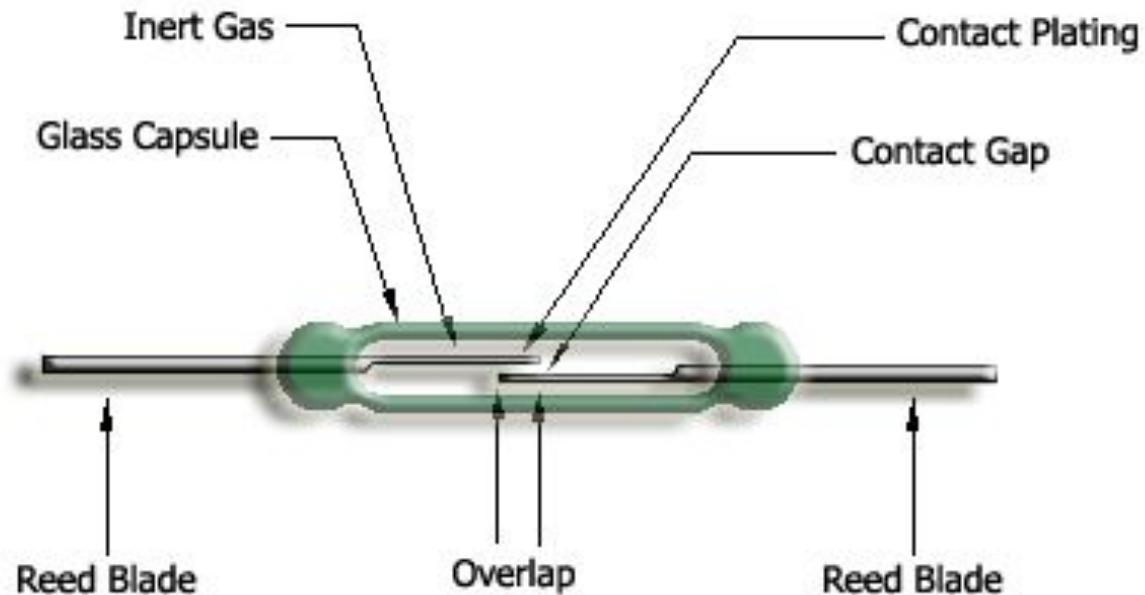
Sensors: Proximity: Inductive

- Similar to Capacitive but requires metallic target
- Either analog or binary output
- Range: 1 – 60 mm
- You might have seen them in 3d printers



Sensors: Proximity: Magnetic

- Hall Effect Sensor
- Reed Switch
- Binary output
- Range: 1 – 5 mm
- Used in some types of motors



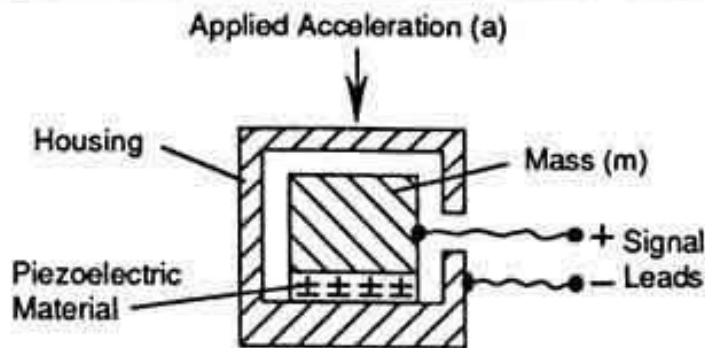
Sensors: Proximity: LIDAR

- Basically a spinning laser sensor (which is similar to infrared)
- Useful for environment mapping
- And for speed traps
- Range: 5 cm – 100 m



Sensors: Orientation

- IMU – Inertial Measurement Unit
- Measures Accelerations
- Can integrate to get position
- Picks up a lot of background noise

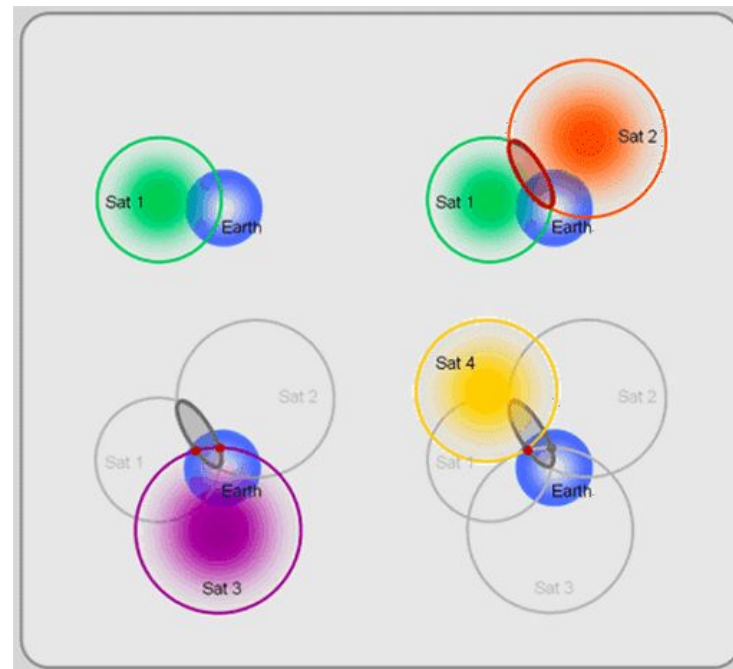
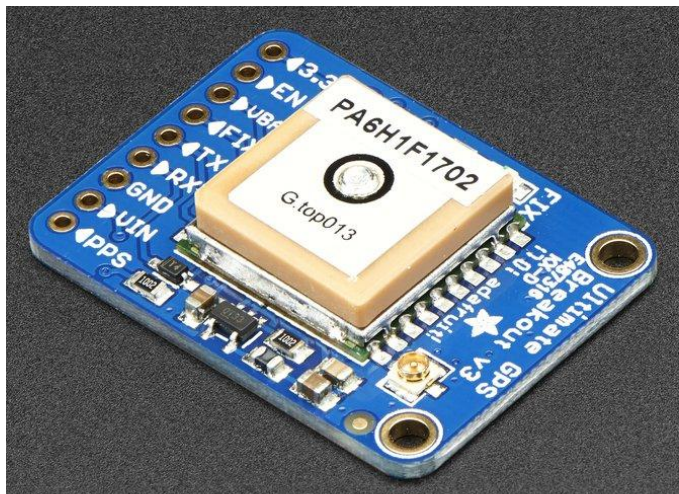


IMUs in Space!



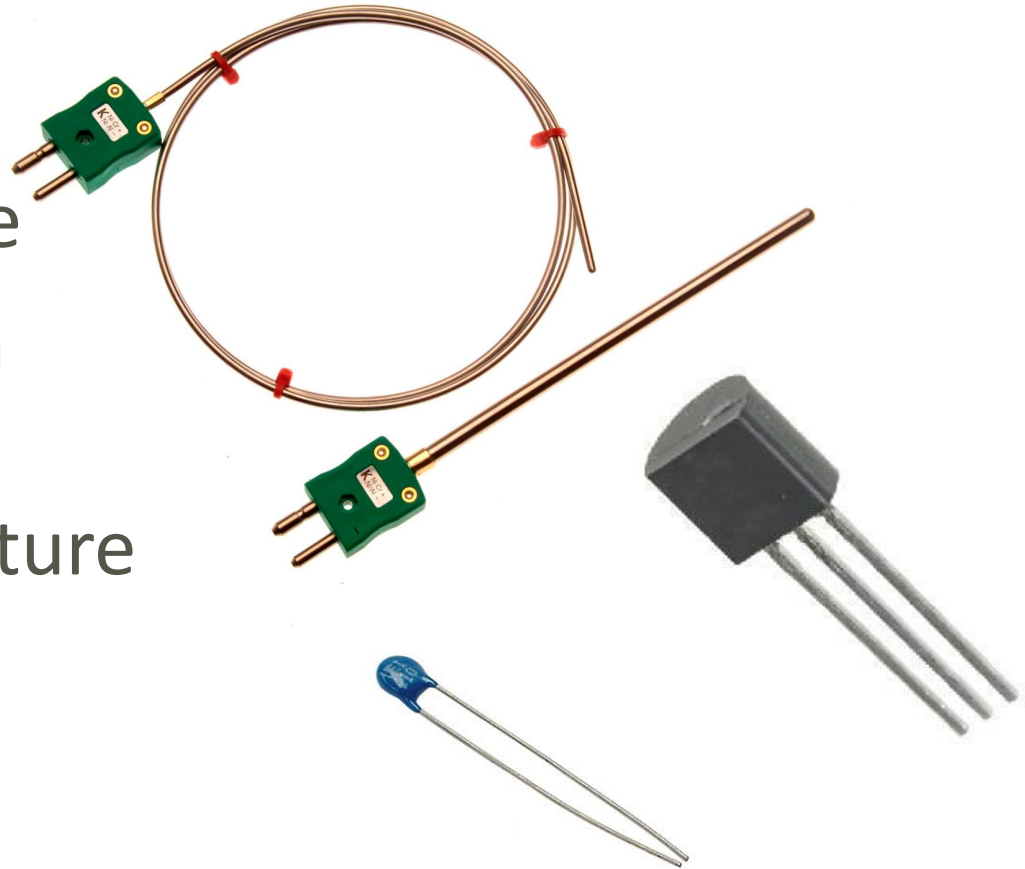
Sensors: Location: GPS

- Location
- Uses distances from 4+ satellites to determine position
- Breakout boards available, but a bit expensive.



Sensors: Temperature

- Thermocouple
 - Voltage difference from differences in two metals
- Resistance Temperature Detector (RTD)
- Thermistor
- Used to monitor temperature



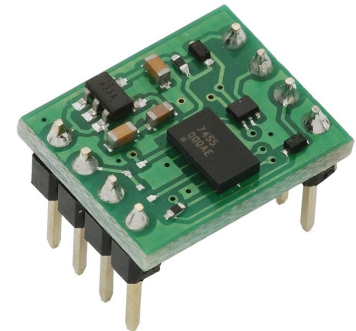
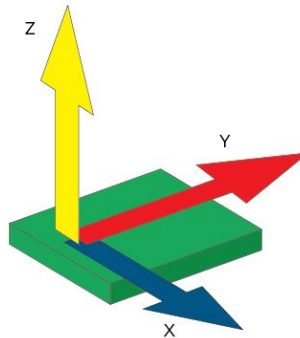
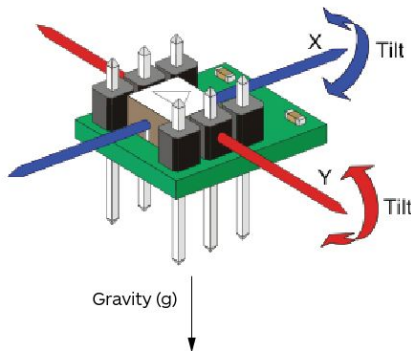
Sensors: Sound Level

- Microphones?
 - Sure!
- Either return soundwave or average sound level
- Respond to changes in air pressure



Your Sensors: Accelerometers

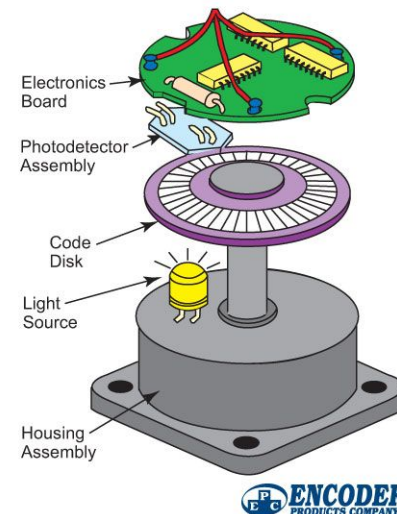
- Measures acceleration.
- Since you can measure acceleration due to gravity, you can determine if your robot is tilted in respect to the Earth.
- They either contain crystal structures sensitive to accelerative forces or capacitance detectors.



Diagrams illustrating the axes of 2-axis (left) and 3-axis accelerometers. This particular 2-axis sensor is also capable of tilt measurement.
Image credit: Parallax | Kerry Wong

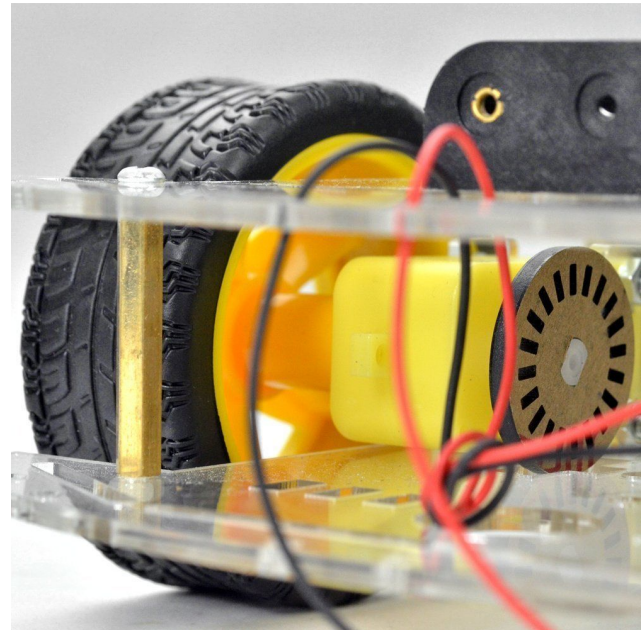
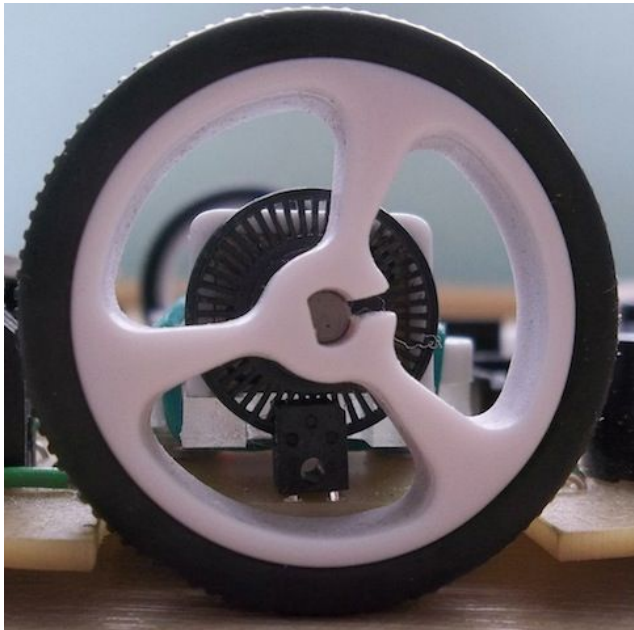
Your Sensors: Encoders

- Converts the motion or position of an actuator to a signal that we can read.
- Lets us know how many times a motor has rotated/ helps us determine distance moved



Bonus Question!

- Why are encoders more accurate than motor timing?
- Why might encoders not be accurate?



Answer:

- Encoders will work better than motor timing if one motor is worse than the other, or if battery power is an issue. In those circumstances, the motor power may not be reliable.
- If the wheel slips, the encoder will register wheel turns without the robot actually moving forward.

